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F. No. 3-6/2019-20/IPM-Advisory
भारत सरकार/Government of India
कृषि एवं किसान कल्याण मंत्रालय/Ministry of Agriculture & Farmers Welfare
कृषि सहकारिता एवं किसान कल्याण विभाग /Department of Agriculture, Cooperation & Farmers Welfare
वनस्पति संरक्षण, संगरोध एवं संग्रह निदेशालय/ Directorate Of Plant Protection, Quarantine And Storage
एन.एच.-4, फरीदाबाद, हरियाणा / N.H.IV. FARIDABAD (HARYANA)

Date: 16 January, 2020

To

1. **The Director of Agriculture,**
   Madan Mohan Malviya Marg,
   Gokhale Vihar, Hazratganj,
   Lucknow, Uttar Pradesh 226001

2. **The Director of Agriculture**
   Government of Haryana,
   Directorate of Agriculture,
   Krishi Bhawan, Sector-21,
   Panchkula, Haryana 134112

3. **The Director of Agriculture**
   Government of Punjab,
   Department of Agriculture,
   Kheti Bhawan,
   Institutional site No.204
   Phase 6, **Mohali**, Punjab

Sub:- Advisory on Late Blight disease of Potato-reg.

Late Blight is a common disease of Potato caused by the fungus *Phytophthora infestans.* The disease affects all plant parts and causes characteristic leaf spots and blight. Low temperature together with saturated atmosphere is congenial for development of Late Blight disease. In view of the prevailing congenial weather conditions, the chances of disease development is high and pest management interventions are imperative to check the possible development of the disease.

The following practices of IPM strategies may be followed for management of Late Blight disease in Potato crop:

- Collect and destroy infested crop debris.
- Maintain field sanitation.
- A strong and foolproof quality control system for fertilizers, bio-fertilizers, fungicides needs to be enforced.

- Regular training programme for the extension workers, pesticide dealers and farmers should be organized by the SAUs/ICAR Institutes/CIPMCs/KVKs for up-gradation of their knowledge and skills regarding management of Late blight.
- Fungicide must be used as per label claim approved by the Central Insecticide Board and Registration Committee (CIB&RC). Detailed information is available at http://ppqs.gov.in/divisions/cibrc/major-uses-of-pesticides. However, list of recommended fungicides for Late Blight of Potato is attached (Annexure-I).
- Farmers should be advised through extension functionaries to use only recommended doses of fertilizers and fungicides as per approved label claims by CIB&RC.
- The IPM package of practices for Potato needs to be promoted. The IPM package of practices for Potato may be downloaded from web-link http://ppqs.gov.in/sites/default/files/potato.pdf

In view of the congenial weather conditions, it is advised to undertake monitoring of potato crop on regular basis. If any Late Blight disease is reported by the farmers, survey of the affected area may be conducted immediately and the concerned officials at district/division/block may be advised to take need based plant protection measures to avoid economic loss. The hot spots of Late Blight disease should be identified through survey for regular monitoring. The detailed information may also be sent to the Directorate through e-mail appa.ipm-ppqs@gov.in urgently to apprise the same to Department of Agriculture, Cooperation and Farmers Welfare.

Encl: As above

(Rajesh Malik)
Plant Protection Adviser

Copy for information:

1. Additional Secretary (PP), MoA& FW, Krishi Bhawan, New Delhi-110001
2. Joint Secretary (PP), DAC&FW, Krishi Bhawan, New Delhi-110001
3. PPS to PPA
4. Officer-In-Charges of RCIPMC Faridabad & Lucknow, CIPMC Jalandhar

Annexure-I

Approved uses of registered pesticides for Late blight of Potato

Crop	Common name of the disease	Dosage per ha			Waiting perio d fromlast applicationto harvest(in days)
		a.i. (g)	Formulation (g/ml)/%	Dilutionin water(L)	
Azoxystrobin 23% SC					
Potato	Late Blight	125 gm	500 ml	500	12
Captan 50% WG					
Potato	Early blight & Late blight	750gm	1500gm	500	21
Captan 50% WP					
Potato	Early blight & Late blight	1250gm	2.5kg	750-1000	-
Captan 75% WP					
Potato	Early blight & Late blight	1250gm	1667gm	1000	8
Chlorothalonil 75% WP					
Potato	Early blight & Late blight	0.66 - 0.937 gm	0.875-1.250 gm	600-800	14
Copper oxy chloride 50% WP					
Potato	Late Blight	1.25	2.5	750-1000	-
Copper Hydroxide 53.8% DF					
Potato	Late blight	525	1500	500	22
Cyazafamid 34.5% SC					
Potato	Late blight	80g	200ml	500	27
Kresoxim-methyl 44.3% SC					
Potato	Early blight & Late blight	250	500	500	23
Mandipropamid 23.4% SC					
Potato	Late blight	0.2 ml/lit	0.8 ml/lit	500-750	40
Oxathiapiprolin 10.1% W/W OD					
Potato	Late blight	20	200	500	22
Ametoctradin + Dimethomorph 20.27% w/w SC					
Potato	Late Blight	420-525	800-1000ml	500	32
Captan70%+ Hexaconazole5%WP					
Potato	Earlyblight	375-750	500-1000	500	21
Carbendazim12%+ Mancozeb 63%WP					
Potato	Early blight, Late blight	210+1102.5			
Carbendazim 25%+ Mancozeb 50% WS					

Potato	Late blight	(1.5 + 3.0) To (1.75 + 3.5) (for 10 kg seed)	6 - 7	2	This is used as seed treatment
Famoxadone16.6%+ Cymoxanil22.1%SC					
Potato	Late blight	210	500	500	40
Kresoxim-methyl 15% + Chlorothalonil 56% WG					
Potato	Early blight and Late blight	Kresoxim-methyl-150 & Chlorothalonil-560	1000	500	23
Metalaxyl M 3.3%+ Chlorothalonil 33.1% SC					
Potato	Late blight	0.073%	0.2%	500	34
Metiram 55% + Pyraclostrobin 5% WG					
Potato	Late blight	900-1050	1500-1750	500	15
Dimethomorph50%WP					
Potato	Late blight	500gm	1000gm	750L	16
Hexaconazole2%SC					
Potato	Early blight& Late blight	60gm	3.0L	500	21
Mancozeb75%WP					
Potato	Late blight	1.125-1.5 kg	1.5-2kg	750Lt	
Metiram70%WG					
Potato	Late blight	1400gm	2000gm	500-750Lt.	21
Propineb70%WP					
Potato	Early& Late Blight	0.21% or 210 G/100Lt. water	0.30% or 300 Gram/100Lt. water	As required depending upon crop stage and plant protection equipment used	15
Zineb75%WP					
Potato	Late blight	1.125-1.5KG	1.5-2KG	750-1000 Lt	
Carbendazim 1.92% + Mancozeb 10.08% GR					
	Late blight	210+1102.5			
Metalaxyl M4%+ Mancozeb64%WP					
Potato	Late blight	0.17%or 1700 gm	0.25% or 2500gm	500-1000 lt.	24
Metalaxyl 8%+ Mancozeb64%WP					

Potato	Late blight	1800 gm or 0.18%	2500gmor 0.25%	1000lt.	Not less Than 7 weeks